

Deployment-Aware Evaluation of TabTransformer for Bank Term Deposit Subscription Prediction

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ABSTRACT

Accurate prediction of customer subscription to bank term deposits is essential for optimizing financial marketing strategies and reducing operational costs. While classical machine learning models have traditionally dominated structured tabular learning tasks, transformer-based architectures have recently emerged as promising alternatives due to their ability to model contextual feature interactions using self-attention mechanisms.

This study investigates the performance of a tuned TabTransformer model on the Bank Marketing dataset under two explicitly defined deployment-aware scenarios: (1) post-contact prediction, where interaction-derived variables such as call duration are available, and (2) pre-contact prediction, where such variables are excluded to simulate realistic marketing decision-making prior to customer interaction.

Following standardized preprocessing and stratified data splitting, models were evaluated using Accuracy, Macro-F1, AUC, and Average Precision. In the post-contact setting, the tuned TabTransformer achieved superior ranking performance (AUC = 0.9177), significantly outperforming Logistic Regression and Decision Tree baselines. However, under the pre-contact setting, performance decreased (AUC $\approx$ 0.77), highlighting the predictive dominance of post-interaction variables and the importance of deployment-aware evaluation in financial machine learning systems.

The findings demonstrate that transformer-based architecture effectively capture categorical feature interactions in structured financial datasets while emphasizing the necessity of clearly defining prediction timing to avoid optimistic performance estimation.

KEYWORDS

TabTransformer, Bank Marketing Dataset, Binary Classification, Transformer-Based Models, Deployment-Aware Evaluation, Financial Machine Learning, Data Leakage

1. INTRODUCTION

Predicting customer subscription behavior plays a critical role in financial marketing analytics. Accurate predictive systems enable banks to allocate outreach resources efficiently, reduce unnecessary contact attempts, and improve campaign conversion rates.

Traditional machine learning models such as Logistic Regression and Decision Trees have long been applied to structured financial datasets due to their interpretability and computational efficiency. However, these models often struggle to capture nonlinear interactions among heterogeneous categorical and numerical features.

Recent advances in transformer-based architecture extend attention mechanisms beyond natural language processing into structured tabular data modeling. The TabTransformer architecture leverages contextual embeddings and multi-head self-attention to model complex inter-feature dependencies without imposing artificial ordinal structure.

A key methodological challenge in financial prediction concerns deployment timing. Certain features may only become available after customer interaction. If such features are included without explicitly defining prediction timing, performance metrics may become inflated.

To address this issue, this study evaluates TabTransformer under two clearly defined experimental settings:

1. Post-Contact Prediction (including duration feature)
2. Pre-Contact Prediction (excluding duration feature)

This dual-setting evaluation ensures transparency and methodological rigor.

2. RELATED WORK

Transformer-based architecture has recently emerged as an effective approach for structured tabular learning. A notable contribution in this direction is the TabTransformer model proposed by Huang et al. [1], which applies multi-head self-attention to learned categorical embeddings. By contextualizing categorical features before classification, the architecture enables richer representation learning and demonstrates competitive

performance compared to traditional machine learning baselines across multiple benchmark datasets.

Building upon such developments, Badaro et al. [2] provides a comprehensive survey of transformer-based approaches for tabular data representation. The survey categorizes architectural adaptations designed to capture structural dependencies across heterogeneous feature types and highlights the growing research interest in extending attention mechanisms beyond sequential text modeling to structured data domains.

Recent work further explores multimodal extensions of transformer architecture. Bonnier [3] investigates models that integrate structured tabular inputs with textual fields, illustrating the flexibility of attention-based mechanisms when handling heterogeneous data sources within a unified modeling framework. In addition, Haddad-Neto et al. [4] analyzes data-efficient transformer-based models for tabular classification tasks. Their findings suggest that while transformer architectures can achieve strong predictive performance, results remain sensitive to dataset scale and hyperparameter configuration, particularly under limited data conditions.

A broader empirical evaluation conducted by Ye et al. [5] compares deep learning methods against established tabular baselines across multiple datasets. Their analysis indicates that gradient boosting frameworks continue to serve as highly competitive baselines, especially for medium-sized structured datasets. These results underscore the importance of rigorous comparative evaluation when introducing transformer-based architectures for tabular prediction tasks.

The theoretical foundation underlying these models originates from the self-attention mechanism introduced by Vaswani et al. [6], where multi-head attention enables parallel modeling of complex feature interactions. This mechanism forms the backbone of modern transformer-based architecture applied to structured and semi-structured data.

While transformer-based models provide contextual feature modeling capabilities, their superiority over classical tabular methods is not universally guaranteed. Empirical evidence reported in recent comparative studies [4], [5] suggests that architectural expressiveness alone does not ensure performance gains, and outcomes remain dataset dependent. These observations highlight the necessity of fair baseline comparison and careful experimental design when evaluating transformer-based models for structured financial prediction tasks.

3. Dataset and Data Challenges

The Bank Marketing dataset consists of 11,162 instances with 16 input features, including numerical variables (age, balance, duration) and categorical attributes (job, marital status, contact type). The target variable is binary, indicating whether a client has subscribed to a term deposit.

Challenges include:

- Skewed numerical distributions (balance, duration)
- Mild class imbalance
- Heterogeneous categorical cardinalities
- Potential predictive dominance of post-interaction features
- Additionally, the choice of categorical encoding strategy may influence the performance of linear models such as Logistic Regression.

Preprocessing steps:

- Label encoding for categorical variables
- Standardization of numerical features
- Stratified 70/15/15 train-validation-test split

Categorical variables were encoded using label encoding for transformer-based modeling, as embeddings are learned during training. However, it is important to note that for linear models such as Logistic Regression, one-hot encoding may provide a more appropriate representation by avoiding artificial ordinal assumptions. This may partially influence baseline performance comparisons.

4. Methodology

4.1 Problem Formulation

Given a feature vector $x_i \in \mathbb{R}^d$ and binary label $y_i \in \{0,1\}$, the objective is to learn:

$$\hat{y}_i = f(x_i; \theta)$$

Training minimizes Binary Cross-Entropy:

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)$$

4.2 TabTransformer Architecture

Categorical features are embedded into dense vectors of dimension d_{model} . These embeddings are processed by stacked Transformer Encoder blocks consisting of:

- Multi-Head Self-Attention
- Position-wise Feed-Forward Networks
- Residual Connections
- Layer Normalization

Numerical features bypass the attention mechanism and are concatenated prior to final classification.

The model was implemented using the Keras deep learning framework.

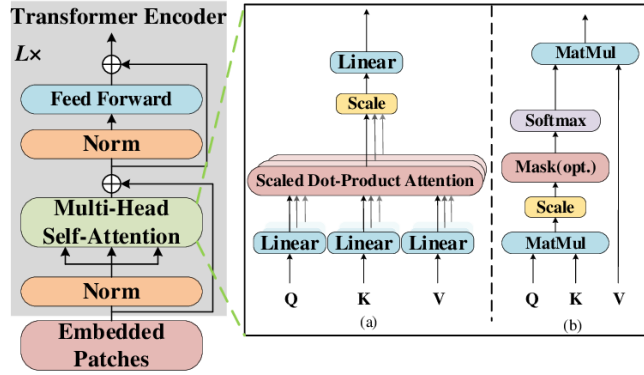


Figure 1. Overview of the TabTransformer model integrating categorical embedding processes through Transformer encoder blocks with numerical features for final classification adapted from Vaswani et al. (2017)

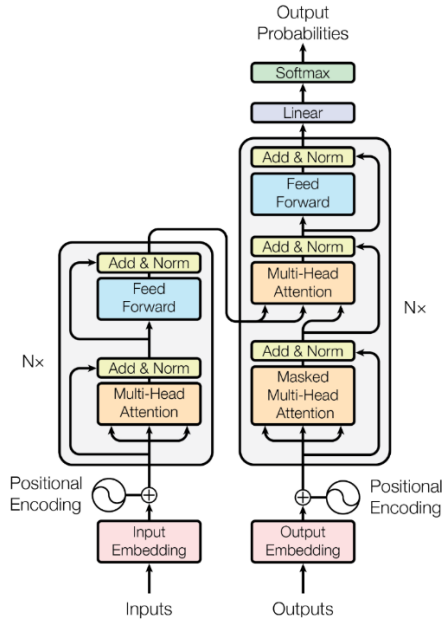


Figure 2. Transformer encoder block structure consisting of multi-head self-attention and feed-forward sublayers with residual connections and normalization [6].

4.3 Hyperparameter Configuration

Default configuration:

- dim = 32
- heads = 4
- blocks = 2
- ff_dim = 128

Hyperparameter search explored:

- dim ∈ {16, 32, 64}
- heads ∈ {2, 4, 8}

- blocks ∈ {1, 2, 3}
- ff_dim ∈ {32, 64, 128}

Final selection was based on validation AUC.

5. EXPERIMENTS

5.1 Data Splitting and Preprocessing

Experiments were conducted using a stratified 70/15/15 split. Numerical scaling parameters were computed exclusively from the training set.

Training utilized the Adam optimizer with learning rate 1e-3 and batch size 32. Early stopping was applied based on validation loss with patience of 5 epochs.

Evaluation metrics included Accuracy, Macro-F1, AUC, and Average Precision.

5.2 Baseline Models

Logistic Regression and Decision Tree models were implemented using scikit-learn under identical preprocessing conditions. Hyperparameters were set to default configurations to provide fair baseline comparison. All baseline models were trained and evaluated using the same stratified data split and performance metrics as the TabTransformer.

5.3 TabTransformer Training Configuration

The TabTransformer model was implemented using TensorFlow/Keras. The tuned configuration consisted of:

- dim = 32
- heads = 4
- blocks = 2
- ff_dim = 128
- batch_size = 32
- maximum epochs = 15

Training was performed using the Adam optimizer with Binary Cross-Entropy loss. Early stopping was applied based on validation loss to prevent overfitting, and the best validation weights were restored.

All hyperparameters were selected through validation-based tuning.

6. Results and Discussion

This section presents model performance under two explicitly defined deployment scenarios: post-contact prediction (including the duration feature) and pre-contact prediction (excluding duration). Each setting is evaluated using identical preprocessing, splitting, and evaluation metrics to ensure fair comparison.

6.1 Post-Contact Prediction (With Duration)

This setting includes the duration feature, which reflects customer interaction behavior during the call. Therefore, this configuration simulates a post-contact evaluation scenario.

Table 1. Overall performance of baseline models.

Model	Accuracy	AUC	AP	Macro-F1
Logistic Regression	0.7922	0.8610	0.8315	~0.79
Decision Tree	0.7875	0.8463	0.7752	~0.79
TabTransformer	0.8400	0.9177	0.8888	~0.84

The tuned TabTransformer achieves the highest discriminative performance (AUC = 0.9177), significantly outperforming classical baselines. The improvement in Average Precision (0.8888) further indicates enhanced ranking capability for the positive class.

The inclusion of duration substantially enhances predictive capability across all models, confirming that interaction-derived variables encode strong behavioral signals.

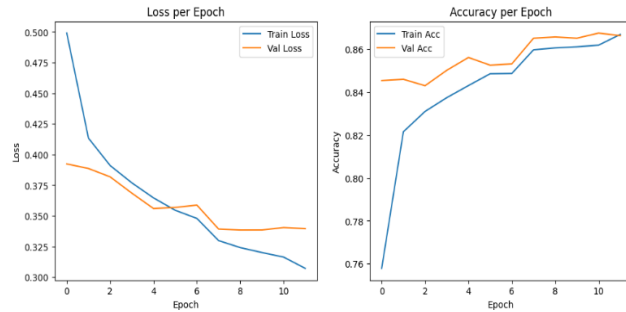


Figure 3. Training and validation loss and accuracy curves for the tuned TabTransformer under post-contact prediction. The curves indicate stable convergence without severe overfitting.

The training curves demonstrate controlled optimization. Early stopping limits overfitting, and the small generalization gap suggests effective regularization.

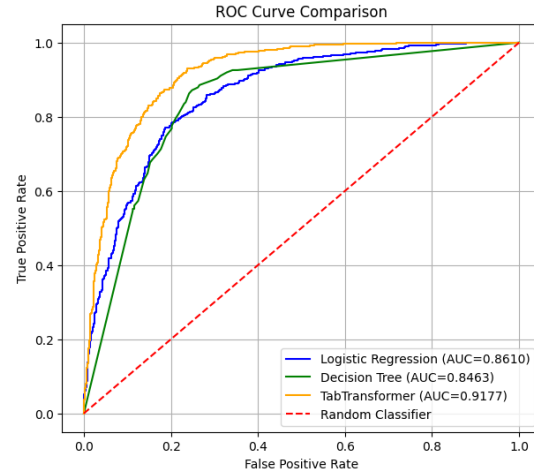


Figure 4. ROC curve comparison across Logistic Regression, Decision Tree, and TabTransformer under post-contact evaluation. The transformer model maintains consistently higher true positive rates across thresholds.

The TabTransformer curve dominates across nearly all threshold values, demonstrating improved ranking stability and probability calibration.

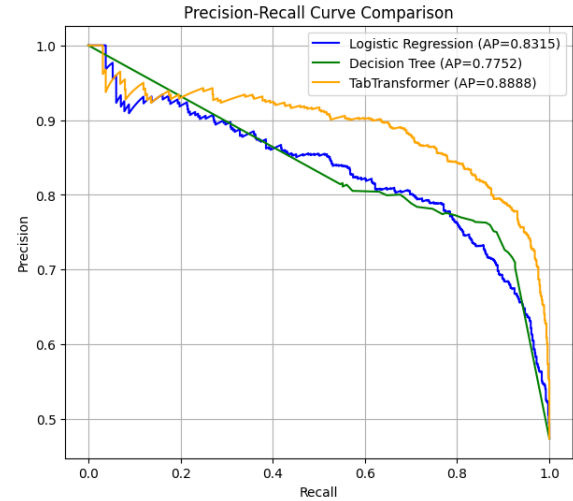


Figure 5. Precision–Recall curves under post-contact setting. The TabTransformer achieves the highest Average Precision, indicating improved positive-class detection under mild class imbalance.

Given the mild class imbalance, the PR curve provides additional insight into positive-class detection. The transformer model maintains higher precision across recall levels.

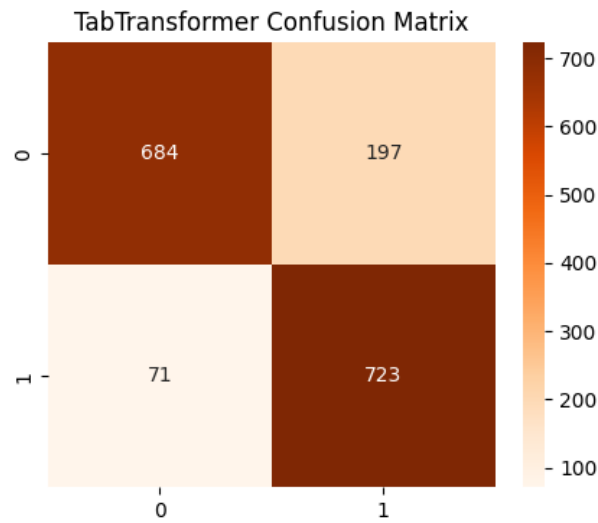


Figure 6. Confusion matrix of the tuned TabTransformer under the post-contact prediction setting (decision threshold = 0.5).

To further analyze classification behavior at the default decision threshold (0.5), the confusion matrix is presented in Figure 6. The model achieves a high number of true positives and true negatives, indicating strong discriminative ability when interaction-derived features are included. False positives and false negatives remain relatively limited, consistent with the high AUC and Average Precision observed in this setting. The balanced distribution of classification outcomes suggests that the transformer model effectively leverages contextual feature interactions under post-contact evaluation.

6.2 Pre-Contact Prediction (Without Duration)

To simulate a realistic marketing deployment scenario, the duration feature was removed. This setting reflects decision-making prior to customer interaction.

Table 2. Model Performance (without Duration)

Model	Accuracy	AUC	AP	Macro-F1
Logistic Regression	~0.66	0.7174	0.7010	~0.66
Decision Tree	~0.65	0.6871	0.6540	~0.64
TabTransformer	~0.72	0.7713	0.7587	~0.71

Removing duration reduces performance across all models. The TabTransformer experiences a decrease of approximately 0.15 AUC compared to the post-contact setting. Despite this reduction, the transformer model remains superior to classical baselines, suggesting that contextual embeddings effectively capture inter-feature dependencies even in the absence of highly dominant features.

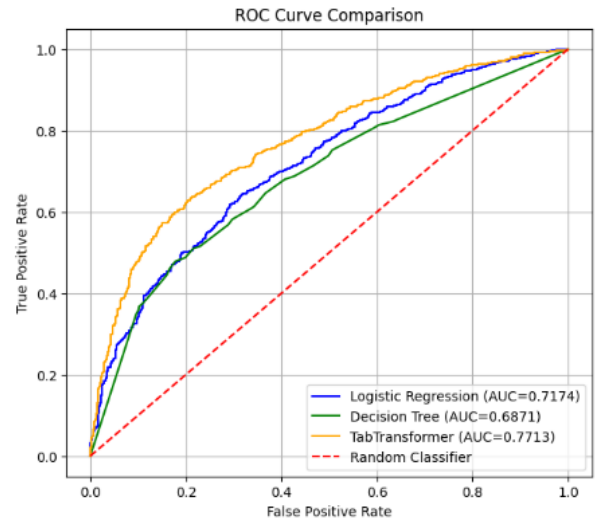


Figure 7. ROC curve comparison under pre-contact setting. Performance decreases across models after removing duration, but TabTransformer remains dominant.

The consistent reduction across models confirms that the performance gap is driven by feature removal rather than architectural overfitting.

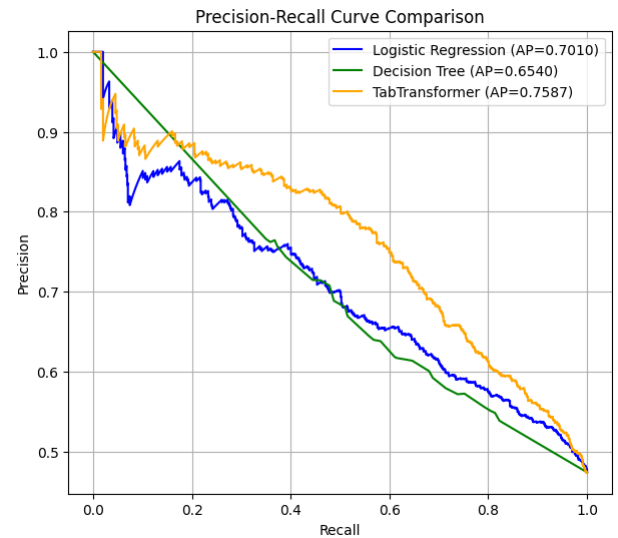


Figure 8. Precision-Recall curves under pre-contact prediction. The transformer model preserves higher precision compared to baselines despite removal of interaction features.

Although absolute performance decreases, the transformer architecture maintains stronger probability ranking under deployment-constrained conditions.

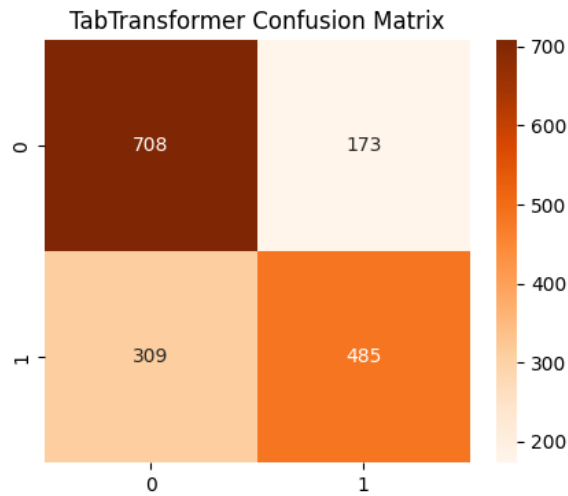


Figure 9. Confusion matrix of the tuned TabTransformer under the pre-contact prediction setting (decision threshold = 0.5).

Figure 9 illustrates the classification behavior after removing the duration feature. A noticeable increase in false negatives and false positives is observed compared to the post-contact setting. This performance shift reflects the removal of a highly informative interaction-derived variable.

Despite the reduction in absolute performance, the model maintains reasonable classification balance, demonstrating its ability to capture meaningful patterns even under deployment-constrained conditions.

6.3 Comparative Deployment Analysis

Table 3. AUC Comparison Across Settings

Model	With Duration	Without Duration
Logistic Regression	0.8610	0.7174
Decision Tree	0.8463	0.6871
TabTransformer	0.9177	0.7713

The substantial performance gap highlights the importance of explicitly defining prediction timing.

- Post-contact prediction yields high discriminative power but reflects interaction-aware evaluation.
- Pre-contact prediction provides a more realistic deployment scenario for proactive marketing strategies.

This distinction prevents optimistic performance interpretation and ensures methodological clarity.

6.4 Model Behavior and Generalization

Training dynamics indicate stable convergence with no severe overfitting. Dropout regularization and early stopping contribute to controlled optimization.

While transformer-based architecture demonstrates strong representation learning capability, they introduce higher computational cost compared to linear models. For medium-sized tabular datasets, careful hyperparameter tuning remains essential to balance expressiveness and generalization.

7. Conclusion

This study investigated the effectiveness of a tuned TabTransformer model for bank term deposit subscription prediction under two deployment-aware scenarios: post-contact and pre-contact evaluation. Results demonstrate that the transformer-based architecture consistently outperformed Logistic Regression and Decision Tree baselines in ranking-based metrics, particularly when interaction-derived features such as call duration were included.

The comparative analysis between post-contact and pre-contact settings revealed a substantial performance gap, emphasizing the predictive dominance of interaction-based variables and the importance of clearly defining prediction timing. While the inclusion of duration significantly improves classification performance, pre-contact evaluation provides a more realistic assessment for proactive marketing deployment. These findings highlight the capacity of attention-based models to capture complex categorical feature interactions in structured financial datasets. However, transformer architecture requires careful hyperparameter tuning and introduces higher computational cost compared to linear models. Additionally, the categorical encoding strategy may influence baseline performance, particularly for linear classifiers.

Overall, this study underscores the necessity of deployment-aware evaluation in financial machine learning systems and provides empirical evidence supporting the applicability of transformer-based models in structured tabular domains.

8. Limitations and Future Work

Despite the strong empirical performance of the TabTransformer model, several limitations should be acknowledged.

First, the dataset size is moderate, which may limit the full expressive potential of transformer-based architectures. While attention mechanisms effectively capture categorical feature interactions, larger-scale structured datasets may better leverage the representational capacity of such models.

Second, a unified preprocessing and encoding pipeline was adopted across all models to ensure methodological consistency. However, categorical encoding strategies can influence model-specific performance. Linear classifiers such

as Logistic Regression may benefit from one-hot encoding, which avoids artificial ordinal assumptions introduced by label encoding. Although this approach ensured fair pipeline consistency, future studies may explore model-specific encoding strategies to further refine baseline comparisons.

Third, the present study primarily focused on predictive performance metrics without extensive interpretability analysis. In financial decision-making systems, explainability remains essential for regulatory compliance and stakeholder trust. Incorporating feature attribution techniques such as SHAP or attention-based interpretation methods could enhance model transparency.

Future research may also investigate alternative regularization strategies, cross-validation schemes, and calibration analysis to improve robustness under deployment conditions. Additionally, comparative evaluation against gradient boosting frameworks such as XGBoost or LightGBM may provide further insight into the relative strengths of transformer-based architectures for medium-sized tabular datasets.

Overall, addressing these limitations would strengthen methodological rigor and support practical deployment of transformer-based models in financial systems.

9. References

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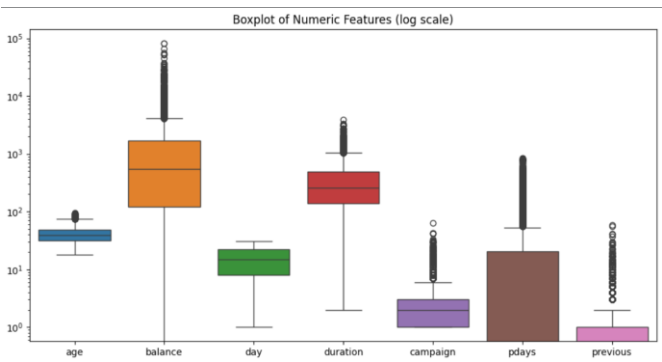
10. Appendix A: Exploratory Data Analysis (EDA) Figures

This appendix presents supplementary exploratory data analysis (EDA) figures generated during the preprocessing stage. These visualizations were used to inspect feature distributions, identify outliers, and understand categorical patterns but were not included in the main body of the report to maintain conciseness.

Figure A1. Distribution of the Target Variable (deposit)

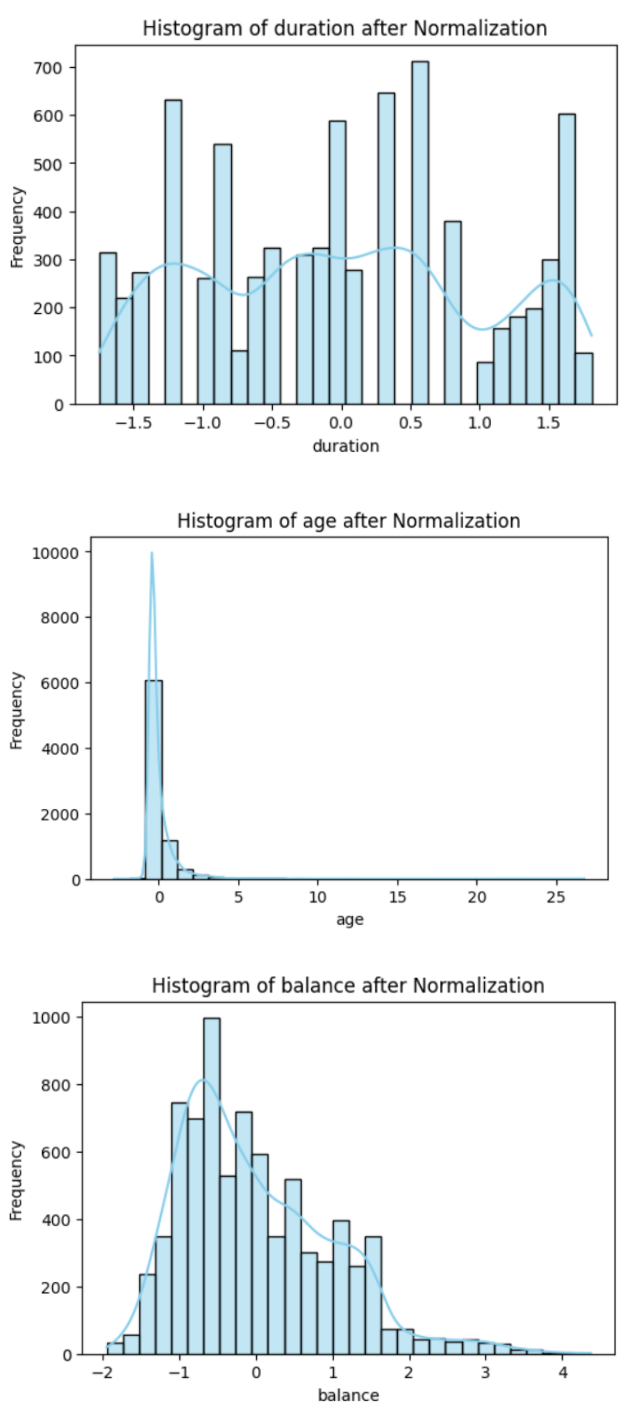


Figure A2. Boxplot of Numerical Features After Standardization



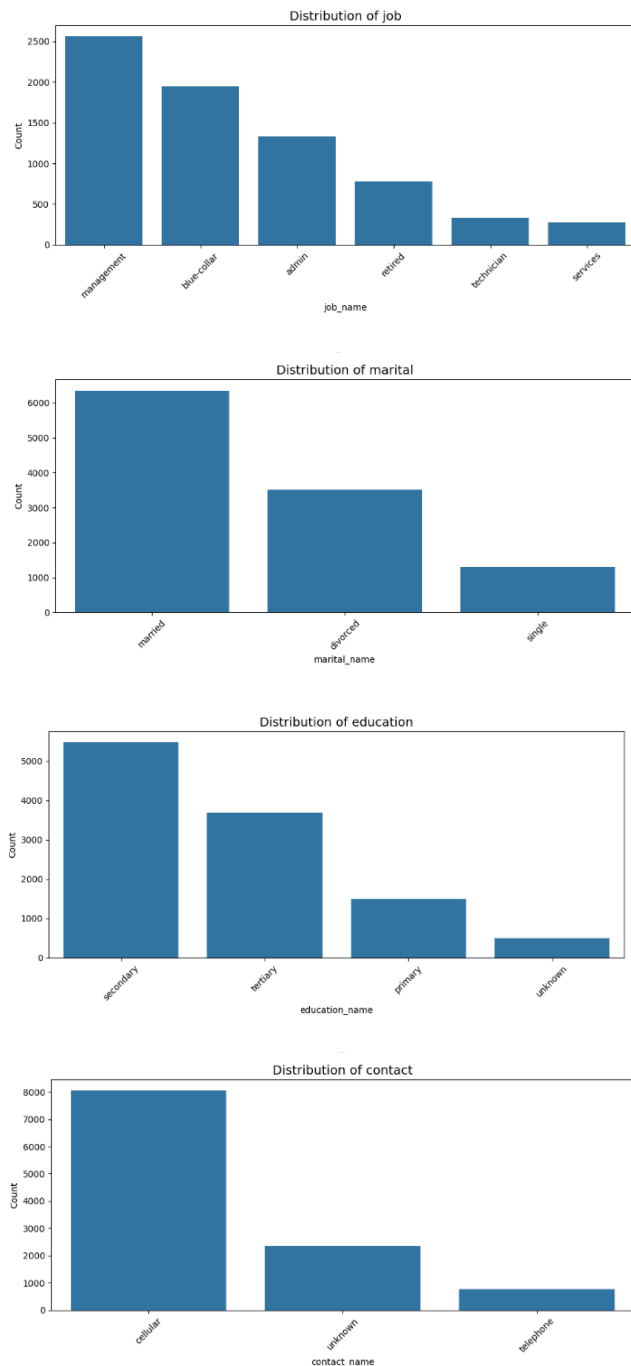
This plot highlights outliers and variability in the numerical features, which may influence model behavior.

Figure A3. Histogram of Numerical Features (Example: age, duration, balance)



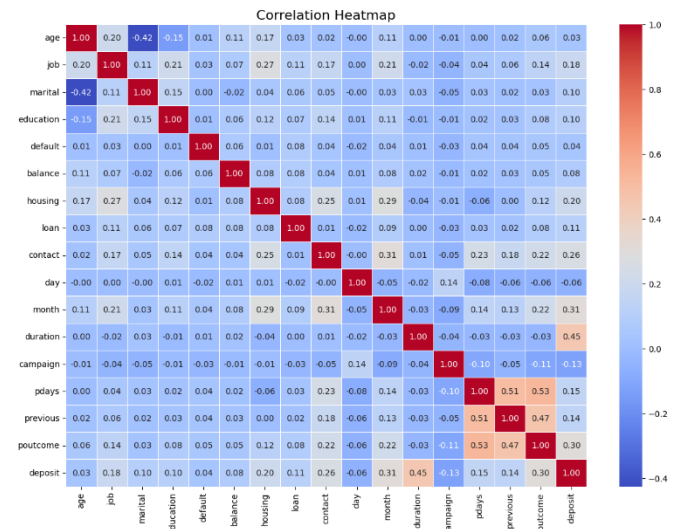
These histograms illustrate the skewed distributions commonly found in financial datasets.

Figure A4. Sample Histogram for a Categorical Feature Mapped to Ordinal IDs (*job*, *marital*, *education*, *contact*, etc.)



Although label encoding was not yet applied during modeling, integer-mapping was used temporarily for visualizing frequency distributions.

Figure A5. Correlation Heatmap Using Target Encoding



The heatmap provides an approximate measure of how features relate to the target variable after applying target encoding for visualization purposes.

